

Example 1. Assume that there is a digit a that makes the following equality true.

$$(512) \cdot (1a53125) = (1000000000)$$

Work modulo 11 to find a .

Solution. Take both sides modulo 11. Observe there are 9 zeroes in 1000000000.

Now solve for a .

So the digit a must be:

Example 2. Let p be a prime and k an integer in $[1 \dots p - 1]$. Prove

$$\binom{p}{k} \equiv 0 \pmod{p}$$

Solution. All binomial coefficients are integers. Ours is given by

$$\binom{p}{k} =$$

Division of positive integers producing an integer can be obtained by

Observe that

On the other hand, because k is in $[1 \dots p - 1]$, it follows that

Therefore, upon division producing the binomial coefficient

And therefore:

□

Example 3. Let n be a positive integer.

Use the binomial theorem and that $7^4 = 2401$ to show 7^{4n} and $1 + 400n$ have the same last three digits.

Solution. They will have the same last three digits if

We write

$$7^{4n} \equiv$$

Next we apply the binomial theorem.

$$\dots \equiv$$

$$\equiv$$

A. **Linear Congruences.** Our next task will be solving **linear congruences**, like

$$3x \equiv 9 \pmod{12}$$

The obvious solution is $x = 3$.

Are there other incongruent solutions?

First $\div 3$ (including the modulus) to obtain the equivalent congruence

This gives us a full list of least residues modulo 12 solving $3x \equiv 9 \pmod{12}$.

whence we consider the congruence solved.

Theorem. Let n be a positive integer. Let a, b be integers with $d = \gcd(a, n)$.

The linear congruence

$$ax \equiv b \pmod{n}$$

has a solution if and only if:

in which case the number of incongruent solutions modulo n is:

Proof. $ax \equiv b \pmod{n}$ having a solution is equivalent to Diophantine

having a solution pair:

From our study of Diophantine equations this occurs if and only if

and, if x_0 is a particular, the solutions x are of the form

This gives us incongruent solutions

Corollary. In the setting of the above theorem, if $a \perp n$ then

$$ax \equiv b \pmod{n}$$

has:

Namely that solution is

Example 4. Solve the linear congruence:

$$28x \equiv 14 \pmod{91}$$

First we find $\gcd$ d of the coefficient $a = 28$ and the modulus $n = 91$.

We then divide $\div d$ to obtain equivalent congruence

$$4x \equiv 2 \pmod{13}$$

We can find 4^{-1} modulo 13 using one of two methods.

Either guess-and-check

or unwind the Euclidean algorithm to write d in terms of a and n

Now, we multiply both sides of the congruence by 4^{-1} to obtain

And so the d least residues of solutions modulo 91 are